

Posture Classifier Streamlit App – User Manual

Complete Guide to Setup and Run the Application

1. Prerequisites

Before you start, ensure you have the following installed on your system:

- Python 3.8 or higher
- pip (Python package installer)
- At least 4GB of RAM (8GB recommended)
- A dataset of posture images (5 classes)

2. Required Dependencies

Install these Python packages using pip:

```
pip install streamlit tensorflow matplotlib numpy pandas scikit-learn
```

Run this command in your terminal or command prompt.

If you have a GPU and want faster training, you can install the GPU version of TensorFlow:

```
pip install tensorflow-gpu
```

4. Virtual Environment

Create virtual environment

```
python -m venv venv
```

Activate it

On Windows:

```
venv\Scripts\activate
```

3. Folder Structure

Your project folder must have the following structure:

```
└─ Your_Project_Folder/
    |
    └─ app.py
    └─ Copy of final project.ipynb (Your original notebook file)
    |
    └─ dataset/
        └─ One arm crossed/
        └─ Arms Crossed High (Chest)/
        └─ Arms Crossed Low (Waist)/
        └─ Arms Open and Neutral/
        └─ Hands in Pockets/
```

4. Prepare Your Dataset

Ensure your dataset meets these requirements:

- Each subfolder name matches exactly the five class names shown above.
- Each subfolder contains at least 50-100 images (300+ recommended).
- Images can be in JPG, JPEG, or PNG format.
- Image size is automatically resized to 128×128 pixels during training.

If your folder names differ, the code will automatically detect them. However, the names above are recommended for clarity.

5. How to Run the Streamlit App

Follow these steps step by step:

1. Open a terminal (Command Prompt, PowerShell, or bash).
2. Navigate to your project folder:

```
cd path/to/your/project
```

3. Install the dependencies (if not already done):

```
pip install streamlit tensorflow matplotlib numpy
```

4. Run the Streamlit app:

```
streamlit run app.py
```

5. Your default web browser will open automatically with the app. If not, open the URL shown in the terminal (usually `http://localhost:8501`).

6. Using the App

Once the app is running, you will see:

- A sidebar with a 'Train Model' button – click it to train a new model (may take several minutes).
- After training, accuracy and loss graphs will appear.
- Upload an image using the file uploader – the app will classify it and display the predicted posture with confidence score.
- The app also shows the top-3 predictions to help you understand the model's output.

7. Troubleshooting Common Issues

Problem	Solution
'streamlit' is not recognized	Run <code>`pip install streamlit`</code> first. If still not found, use <code>`python -m streamlit run app.py`</code> .
Dataset path not found	Make sure the 'dataset' folder is in the same directory as <code>app.py</code> and contains the subfolders.
Out of Memory (OOM)	Reduce <code>batch_size</code> in the code from 16 to 8 or 4, or reduce image size to 96x96.
Low prediction confidence	Train for more epochs (increase epochs to 30) or add more training images.
Graphs not showing	Ensure matplotlib is installed and that training completed successfully.
Model not saving	Check write permissions in the project folder. You can also save manually by adding <code>`model.save('posture_model.keras')`</code> after training.

8. Advanced Tips

- To improve accuracy, use transfer learning (e.g., MobileNetV2) by uncommenting the relevant section in `app.py`.
- If you have a GPU, install ``tensorflow-gpu`` for faster training.
- You can adjust the training parameters (epochs, batch size, image size) directly in the ``train_posture_model`` function.

- The model is saved as `posture\_model.keras` after training. You can reuse it without retraining by placing it in the project folder.

9. Final Notes

This app is designed for educational purposes. It demonstrates a complete computer vision pipeline using a CNN. For production use, consider deploying on a cloud service (e.g., Streamlit Cloud, Hugging Face Spaces) and using a more robust model.

Follow my repo for the complete file and the folder structure as well

Github Repository:

<https://github.com/debug-phantom/CCDEPLRL-FINAL-PROJECT-REPO>